

Matthew Emmanuel T. Labrador

matthewtlabrador@gmail.com • matthewlabrador.vercel.app • github.com/M4tyu633 • linkedin.com/in/matthew-labrador

EDUCATION

University of the Philippines Manila

Manila, PH

BS Computer Science — University Scholar | Expected May 2029

Current Cumulative GWA: 1.0375

PAREF Southridge School

Manila, PH

Senior High School — Class Valedictorian, Gold Medalist | May 2025

EXPERIENCE AND PROJECTS

eGovMed — eGov Hackathon PH, Champion (₱100,000 Grand Prize)

2026

Next.js, Node.js, TypeScript, eGov API Suite, Vercel

- Won the grand prize with Bisaya Hackers for an AI triage system that routes patients before they queue.
- Integrated the national eGov API suite end-to-end: AI triage, messaging, reporting, digital identity, and payments.
- Presented the live product demo to the judging panel.

Diabetic Complication Swarm (GlycoSwarm AI) — AMD Developer Hackathon ACT II, Track 3

2026

Python, LangGraph, FastAPI, Next.js, Ollama, AMD MI300X GPU

- Led an international, cross-timezone team building a multi-agent diabetic complication screening prototype.
- Built a LangGraph StateGraph of four parallel agents, each writing and running its own Python scoring code on NHANES lab data.
- Served live inference with Gemma 4 26B on an AMD MI300X GPU via Ollama, with automatic failover to Fireworks GLM 5.2.
- Independently designed and delivered the live demo and slide deck for the judges.

AI Application & LLM Development

2025–2026

Python, Gemma 3B, Qwen 3, LangChain, LangGraph, Anthropic API

- Deployed open LLMs locally (Gemma 3B, Qwen 3) to study end-to-end setup, inference, and multi-agent performance.
- Built automated summarizer applications on the Anthropic API using RAG and Agent-to-Agent (A2A) architectures.
- Ran structured prompt evaluations across model versions and documented the findings.

UPLB Code Wars

2025–2026

C/C++, Algorithms, Compiler-free Environment

- Solved complex algorithmic problems entirely in Notepad, with no compiler, internet, or AI assistance.

Tumbang Preso (TUMP) — DOST-GameDev

2026

Godot 4, GDScript, Blender, Authoritative-Host Multiplayer

- Built a 4-player online multiplayer take on the Filipino street game: four rounds, one rotating defender against three attackers.
- Networked with an authoritative host, resolving contact by distance host-side so tags and throws agree across all peers.

CHIP-8 Emulator

2026

C/C++, Raylib, System Architecture

- Engineered a robust CHIP-8 emulator from scratch, implementing core CPU instructions and memory management.
- Built a custom visual debugger using Raylib to track system state, registers, and memory for analytical debugging.

Heart Disease Prediction Model

2025

Python, pandas, Google Colab

- Trained a supervised learning model in Python to predict heart disease from patient features end-to-end.
- Cleaned data and performed feature engineering with pandas, handling missing values and categorical variables.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, TypeScript, GDScript, HTML

ML & Data: Feature Engineering, Evaluation Metrics (accuracy, precision, recall, F1), pandas, Prompt Engineering, Model Deployment

Agentic AI & Orchestration: LangChain, LangGraph, FastAPI, Multi-Agent Systems, Ollama

Methods & Tools: Statistical Analysis, Model Evaluation, Google Colab, Git, Linux/CLI, Raylib, Godot, Blender, Next.js, Vercel

Certifications: AMD Multiagent Systems Deployment, CS50 Intro to CS, AI Prompt Engineering, Python, C

Spoken Languages: Filipino (native), English, Latin

LEADERSHIP AND ACTIVITIES

Competitive Debater

2022–2025

Various National Tournaments, Philippines

Quarterfinalist at Ateneo Peace Debate; competed at UP Diliman Debates, XSDC, and ASDC.

Google Developer Group and UP Socomsci

2025–2026

Coordinated cross-functional teams on operations and event setup.

Math Team Captain

2022–2024

PAREF Southridge School, Muntinlupa, PH

Led tutoring sessions and mentored peers for local math competitions.

AWARDS

Champion, eGov Hackathon PH (2026, ₱100,000) | DOST Undergraduate Scholar (2025–Present) | Top 5 Finalist, Olympysics NCR (2025) | 5th Place, Philippine Statistics Quiz NCR (2025)